AP CALCULUS AB - UNIT 1

Guided Notes 1.16: Working with the Intermediate Value Theorem

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we state every hypothesis of the Intermediate Value Theorem?

- State every hypothesis of the Intermediate Value Theorem.
- Use endpoint values to guarantee roots or other target outputs.
- Count minimum distinct solutions from disjoint sign-changing intervals.
- Use auxiliary functions and bisection to localize guaranteed solutions.

Key Knowledge, Vocabulary, and Notation

Closed interval — The function must be continuous on $[a, b]$.

Target bracket — The desired value N must lie between $f(a)$ and $f(b)$.

Conclusion — There exists at least one $c \in (a, b)$ such that $f(c) = N$.

Existence, not uniqueness — IVT guarantees at least one solution, not exactly one.

Auxiliary function — Rewrite $F(x) = G(x)$ as $H(x) = F(x) - G(x) = 0$ and apply IVT to H.

Concept Development and Strategy

1. State continuity, endpoint values, the bracketing inequality, and the existence conclusion.
2. Use disjoint intervals to guarantee distinct solutions.
3. For bisection, retain the half-interval whose endpoint values still bracket the target.

Shared Representation / Data

x	0	1	2	3	4
$f(x)$	3	-1	2	5	-2

Worked Example: Guarantee a root

Show $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

Answer and Reasoning

Let $f(x) = x^3 + x - 1$, continuous on $[0, 1]$. Since $f(0) = -1$ and $f(1) = 1$, 0 lies between the endpoint values. IVT guarantees $c \in (0, 1)$ with $f(c) = 0$.

Worked Example: Nonzero target

If f is continuous on $[2, 5]$, $f(2) = 1$, and $f(5) = 9$, show $f(c) = 6$ for some c.

Answer and Reasoning

6 lies between 1 and 9, so IVT guarantees at least one $c \in (2, 5)$ with $f(c) = 6$.

Worked Example: Intersection

Show $e^{-x} = x$ has a solution in $(0, 1)$.

Answer and Reasoning

Let $H(x) = e^{-x} - x$, continuous on $[0, 1]$. $H(0) = 1 > 0$ and $H(1) = e^{-1} - 1 < 0$, so IVT guarantees a zero and hence an intersection.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. List the three main IVT hypotheses/conclusion components.

2. If f is continuous on $[1, 4]$, $f(1) = -2$, and $f(4) = 5$, what value is guaranteed to occur: -3, 0, or 8?

3. Show $x^2 - 3 = 0$ has a solution in $(1, 2)$.

4. Use the shared table to identify intervals that guarantee a root.

5. Show $\cos x = x$ has a solution in $(0, 1)$.

6. Let f be continuous on $[0, 6]$ with $f(0) = 2$, $f(2) = -1$, $f(4) = 3$, $f(6) = -4$. What minimum number of roots is guaranteed?

7. Prove that any odd-degree real polynomial has at least one real root.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Which condition is required by IVT?

(A) Differentiability on (a, b)
(B) Continuity on $[a, b]$
(C) $f(a) = f(b)$
(D) A unique root

2. If f is continuous, $f(2) = -3$, and $f(5) = 4$, IVT guarantees

(A) exactly one root
(B) at least one root in $(2, 5)$
(C) a root at an endpoint only
(D) no root

3. A calculator gives $f(1.2) = -0.4$ and $f(1.3) = 0.2$ for a continuous f. What is guaranteed?

(A) A zero in $(1.2, 1.3)$
(B) Exactly one zero
(C) A maximum
(D) A discontinuity

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

A continuous function satisfies $f(0) = 0$, $f(1) = 2$, $f(2) = 0$. Does IVT guarantee a point where $f(c) = 1$ in both $(0, 1)$ and $(1, 2)$?